

GALXY — MODULE 5 EXECUTION PLAN

v5 — 2 Backend / 2 Frontend Split

Frontend Member 1 · AIPreviewPanel

Frontend Member 1 — AIPreviewPanel

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Component: AIPreviewPanel

Lives On: Product detail page, alongside the product image gallery

What It Does

The full customer-facing AI Preview experience on the product page: toggling into preview mode, showing loading/success/error states, and prompting regeneration when the configuration goes stale.

Behavior Spec

- Tab toggle: sits alongside the product photo gallery, toggling between "Product Photos" and "AI Preview" tabs. Auto-switches to AI Preview the first time the shopper changes any attribute flagged `affects_ai_preview: true` (read from Module 2's schema via Backend Member 1's category data).
- Loading state: an animated glow scan-line or pulsing gradient placeholder while `POST /api/ai/generate-preview` is in flight — never a generic spinner (per the design system).
- Success state: shows the returned image (`output_image_url`), the disclaimer text ("AI-generated approximation..." — static frontend copy), and a "Regenerate" button that re-fires the same request.
- Rate-limit state: on a 429, read `limit_scope` from the response and show the matching message — "You've used your free previews for now — sign up to keep designing" for scope "session" (guest), or "Daily limit reached, try again tomorrow" for scope "user" (logged-in). Never show a raw error.
- Stale-preview banner: when an `affects_ai_preview: true` attribute changes after a preview already exists, dim the current image slightly and show "Preview may be outdated — Regenerate" — don't auto-fire a new generation.

Backend Dependencies

Everything this component needs comes from Backend Member 1's endpoint and Backend Member 2's rate-limit contract.

```
POST /api/ai/generate-preview
success/cache-hit response includes:
  output_image_url, from_cache, generation_id
429 response includes:
  { "success": false, "data": { "limit_reached": true,
    "limit_scope": "session" | "user" } }
```

- `affects_ai_preview` flag comes from Module 2's category attribute schema (already fetched with the product/category data) — read it directly, don't hardcode a list of attribute names.

Handoff Notes

Watch out: Confirm the exact generation-ID field name with Backend Member 1 before wiring Regenerate — you'll need it to re-fire the same request and, later, to match entries coming back from Frontend Member 2's AIGenerationHistory.

Watch out: The disclaimer text and loading animation are yours to design — the backend does not send copy or animation state, only data.

Source: Galxy_Frontend_Detail_All_Modules.docx (Module 5 — AIPreviewPanel), reconciled against Backend Member 1 & 2's contracts.